

# Heather Leffew, PhD

Behavioral Scientist | Psychological Text Analysis | Experimentation and Research Leadership

*Applying the science of human behavior to language, decisions, and measurable business results.*

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## SUMMARY

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Behavioral science and research leader (PhD, Psychology, Quantitative Predictive Linguistics; MA, Advanced Psychometrics) with more than 15 years building and leading applied research functions in regulated, high-stakes businesses, including FCRA-governed consumer reporting, platform safety for more than 180 million US users, and public-safety hiring. Founded and lead Spokeo's Data Science organization, owned the behavioral measurement roadmap and led 25+ member cross-functional teams at TikTok, and directed three teams with P&L authority at Brower. Partner with CEOs and senior leaders in Product, Engineering, Legal, Compliance, and Sales to turn behavioral evidence into product, policy, and investment decisions measured on the business's own KPIs, with a specialty in psychological text analysis, reading state, intent, and likely behavior in the words people choose.

## EMPLOYMENT HISTORY

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### **Director of Data Science | Spokeo** | Sep 2025 to Present

Founding leader of Spokeo's Data Science organization, working in close partnership with the CEO, the executive team, and senior leaders across Product, Engineering, Legal, Compliance, and Sales to make identity, search, and conversational AI products measurable, auditable, and trustworthy for ~20 million unique monthly users. Set the research roadmap and run a portfolio of concurrent programs from first question to shipped product change; chair the cross-functional working group where Product, Engineering, and Data leadership make evidence-based product decisions; regularly present research initiatives, findings, and the resulting product improvements at company all-hands.

- Built the behavioral research program for SpokeoAI, the company's primary product bet, defining its measures of conversational quality (distress, apology, and task success across 79,020 conversations), and used the findings to reset the product roadmap, shifting engineering investment to conversation quality and making name search the top product fix.
- Instituted offline replay testing of ranking changes before live A/B tests (87% agreement with users' actual next actions), and stopped the phone program's founding hypothesis before build on a measured drop in first-correct results from 85.9% to 31.6%, saving a full experiment cycle.
- Own the AI compliance evidence that Legal and Compliance rely on; designed the adjudication program and confirmed zero FCRA-prohibited screening failures across 204 adjudicated conversations on a live LLM product, and lead company-wide ISO/IEC 42001 AI Management System readiness through influence across every model-owning team.
- Advise the executive team on multi-million-dollar data investments; built the decision case for a \$1.5M+ annual Equifax renewal, set a net-new-value standard for data purchases and sized one purchase at ~49.6M people gaining a deliverable email, and scoped a planned multi-million-dollar rebuild down to a targeted repair by measuring all 59 data sources and isolating the issue to one matching tier (35,000 of 652M entities).
- Established independent precision measurement for identity-linking changes, tied ship decisions to break-even economics at \$0.20 per record, and architected a peer-reviewed deployment gating policy for catching circular-evaluation and construct-validity failures before release.
- Built a company-wide quality signal from the language of 45,983 customer care tickets, hand-coding 1,423 into a 14-class scheme (kappa 0.740), and used monthly alerting to catch a deceased-record wave and a repeat-billing incident within days.
- Won same-day agreement from the CEO and engineering leadership on a move from six-week to four-week release cycles by framing it as additive to their existing work, and built Sales a self-service coverage resource so they answer routine coverage questions on their own.

### **Principal Data Scientist & Applied Researcher | TikTok USDS, ByteDance** | Jul 2022 to Jul 2025

Principal research leader in Trust and Safety for a platform with more than 180 million US users and annual revenue above \$20 billion. Owned the behavioral measurement and detection roadmap across mental health, minor safety, election integrity, violent extremism, and brand safety, leading 25+ member cross-functional teams; advised legal, policy, and executive stakeholders and served as the sole authorized researcher on analyses relied on during Congressional scrutiny.

- Founded the external research program, creating the industry's first external IRB process for platform-scale behavioral research and the lab's first academic partnership framework; opened collaborations with Harvard Medical School (SAMHSA Recovery Month) and the Ontario Brain Institute, and represented the program at TrustCon 2024.
- Set the research agenda on user wellbeing, initiating Project Chiron (the first internal audit of recommendation-feed effects on users with depression and suicidality, later the foundation for the Ontario Brain Institute study) and SearchMD (a multi-study program on medical information trust combining IRB-reviewed UX studies and qualitative creator research).
- Identified the recommendation feed as the early-intervention point for habitual violative consumption (70% of violative views came through For You; users who downloaded content had 19 times the odds of habitual consumption), and gave Trust and Safety a specific surface and predictor to act on.
- Cleared ~23,000 accounts for immediate ban and identified 142,288 more as eligible in minor-safety enforcement, with appeal rates unchanged; built detection across 1B+ records at 95% accuracy and reduced manual moderation review by 40%.
- Established over-enforcement as a measured harm, showing that more than 64% of posts flagged as harmful were support and safety-promoting content; introduced within-subjects, FDR-controlled subgroup-harm testing for product changes, and won adoption by rebuilding a 94%-precision black-box model as an interpretable one.
- Built the threat-assessment practice for violent extremism, including a Risk/Threat Rating Matrix and new law-enforcement escalation protocols (33,706 accounts surfaced for review), and served as primary consultant and trainer on FATE, an award-winning linguistic analysis tool adopted across verticals.
- Built consensus across five teams on election-integrity measurement and contributed analysis to a 60% reduction in viral election misinformation compared with the 2020 cycle.
- Built SCOPEX, a psychological text-analysis framework (factor analysis, topic models, collocation, sentiment) for tracking change in an account's language and consumption over time; applied it across 136,950 observations, and the legal team adopted it for a multi-account mass-casualty matter.
- Partnered with Global Research to build wellbeing measurement for staff exposed to traumatic content, unifying survey instruments into one factor structure tracked by person, region, and role.

#### **Head of Accreditation | National Emergency Responder and Public Safety Center | Nov 2020 to Jul 2022**

Partnered with the executive, sales, product, and content teams to lead accreditation, evaluation, and product operations for a public-safety training and certification organization. Managed the full accreditation lifecycle as we expanded nationally to serve more than 1,000 public safety agencies, leading to a multi-million dollar acquisition offer.

- Built standards-aligned assessment and review frameworks for emergency responder training, certification, renewal, and site-visit processes, scaling from regional to national accreditation coverage.
- Authored winning RFPs translating agency requirements into practical product and evaluation scope; presented risk analyses and accreditation findings to government officials and oversight bodies.
- Authored and scaled critical-incident and threat-assessment curricula across the national certification program.

#### **Director of Evaluations and Analytics | Brower Psychological Police and Public Safety | Jun 2019 to Jul 2022**

Directed evaluations and analytics with P&L authority, leading three cross-functional teams (15 individual contributors, 3 supervisors) that delivered digital psychological assessment, people analytics, and wellness programs to public-safety agencies across three states.

- Grew the testing division from an \$81K annual loss into the organization's primary revenue driver; partnered with executive leadership on the move from paper to digital evaluation and delivered a \$781K profit swing in about 14 months (6.2x revenue, 5.8x throughput, 88% lower cost per evaluation).
- Led the organizational change for the turnaround, redesigning the workforce model (one-month training for retired detectives in place of six-month psychometrist onboarding), doubling the client base to 200+ agencies (~70% regional share), raising NPS from 34 to 72, and securing \$500K+ in grants.
- Launched an analytics product adopted by 100% of agencies processing 100+ candidates a year, and delivered threat assessment, critical-incident response, and training to public-safety agencies.

#### **Consulting Staff Data Scientist | QMS Infotech, Inc. | Mar 2008 to Oct 2019**

Advised the leadership of enterprise and public-sector clients on hiring, workforce assessment, and fairness strategy, partnering with executives to drive organizational change in how they made and governed selection decisions. Built analytics and audit products for private-sector hiring, workforce assessment, fairness review, and FCRA-compliant HR decision support.

- Set the psychometric and statistical standards clients adopted for talent-assessment audits (construct validity, fairness measurement, defensible decision criteria), and led bias-mitigation policy changes for enterprise HR partners, achieving full FCRA compliance.
- Led a multi-agency adverse-impact audit of 1,001 pre-employment psychological evaluations for a police department's consent-decree review; certified three years as bias-free and traced a 2021 deviation to a likely recruiting change.
- Turned audit findings into repeatable governance and remediation workflows that client leadership used to change selection practices across their organizations.

#### **Practicum, Doctoral & Post-Doctoral Positions | Fielding Graduate University | May 2015 to Jul 2019**

Clinical and research leadership across psychiatric, crisis, and public-safety settings (Cedar Springs Psychiatric Hospital, Rocky Mountain Behavioral Health, Brower Psychological). Supervised and trained 12 master's-level clinicians and practicum students in assessment methodology, case conceptualization, and clinical reporting standards, while conducting psychological, neuropsychological, occupational, and forensic assessments, including semi-structured interviews.

- Served as Assistant Disaster Response Coordinator for Aurora Mental Health Center's disaster-response program, built after the 2012 Aurora theater shooting, coordinating on-site behavioral-health response and crisis intervention in mass-casualty events, including the aftermath of the 2019 STEM School Highlands Ranch shooting.
- Supported crisis intervention and fitness-for-duty evaluation for public-safety agencies in high-consequence operational settings.
- Led data preparation and analysis for the Hawkins TAT Project, scoring 69 participants' narratives on about 60 LIWC variables at card, person, and corpus level, and used MANOVA to show distinct emotional and social themes across TAT cards (largest effect  $\eta^2 = .53$ ).

#### **Additional Professional Experience | Mar 2002 to Aug 2015**

- Benefits Analyst, Florida Department of Children and Families. Conducted eligibility determinations for state and federal assistance through applicant interviews, policy interpretation, and fraud identification.
- Professional Compliance, Broward Sheriff's Office. Combined emergency communications with audit, compliance, and risk management; led digital transformation of audit systems.
- Neurotherapist, Unity Recovery Services. Managed neurotherapy delivery for a substance-use recovery program and built intake-to-discharge symptom tracking.
- Consultant, Hope for Families. Strengthened certification, accreditation, audit readiness, and regulatory compliance for private-adoption operations.

## **EDUCATION**

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**Doctor of Philosophy (PhD), Psychology, Quantitative Predictive Linguistics | Fielding Graduate University (APA Accredited)**

**Master of Arts (MA), Psychology, Advanced Psychometrics | Fielding Graduate University (WSCUC Accredited), GPA 4.14**

**Bachelor of Science (BS), Psychology, Emergency Management & Public Safety | Liberty University (SACSCOC Accredited), Summa Cum Laude, GPA 4.00**

## **SELECTED RESEARCH**

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- **Instrumental and Affective Mass Murder: Establishing a Predictive Typology (dissertation).** Clustered 100 pre-attack communications on four LIWC dimensions into affective (n=53) and instrumental (n=47) types, and found affective attackers 7.65 times more likely to end the attack by suicide or apprehension ( $p = .006$ ) and a McFadden  $R^2$  of .24 for a logistic model of attack behavior.
- **LIWC and the TAT.** Led the LIWC analysis of 677 Thematic Apperception Test narratives from 69 participants and showed that each card elicits its expected themes; Card 8BM stories carried 3.18 times the corpus rate of death language.
- **Implicit Power Drives in the Manifestos Preceding Autogenic Massacres.** Found LIWC power scores in 23 manifestos above published norms for blogs, expressive writing, and a multi-source composite ( $t(22) = 3.47, 3.66, \text{ and } 2.43$ ), and used the results to select the dissertation's variables.
- **Complaint Text as an Early-Warning Instrument.** Built a reproducible tutorial on turning complaint language into a monitored early-warning signal, with a written codebook, two-coder reliability, a classifier, a blind validation draw, and an alert backtest, and extended it to attorney representation in claims.

- **The Helper's Language as an Early Indicator.** Built a two-speaker, LIWC-style measure of service conversations that predicts attorney representation from the first contact, corrects for conversation length, and sizes a team-randomized coaching experiment.
- **UX Principles and Livestreamed Violence.** Analyzed eight livestreamed attacks and platform reporting design against Hick's Law, Jakob's Law, and the von Restorff effect, and designed an A/B test of a simpler claim-reporting menu.
- **The AI Productivity J-Curve on the Front Line.** Built an illustrative simulator of the productivity dip when claim teams adopt an AI tool, and designed a stepped-wedge office rollout that measures cycle time and rework.
- **Pathologizing Without Warrant (preprint).** Coded a 26,158-post public corpus against a clinical boundary-violation framework (Gutheil and Gabbard) in a 46-page preprint, documenting 35 cases of an AI model issuing unsolicited psychiatric attributions to its users and escalating in all 5 cases where users pushed back.
- **The Somatic Deficit: Moral Language Without Moral Ontology.** Proposed a theory applying psychopathy and moral-development research (Kohlberg, Lykken) to model alignment, and designed three falsifiable studies to test it.

## SKILLS

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**AI Evaluation & Safety:** AI Evals, LLM-as-Judge, Red-Teaming, Automated Red-Teaming Pipelines, Golden Datasets, Agentic Evals, Long-Horizon Reasoning Evaluation, Tool-Use Evaluation, RAG Evals, Simulation Testing, Hallucination Detection, Jailbreak Resistance Scoring, Prompt Injection Detection, Out-of-Character Behavior Detection, Context-Blindness Measurement, In-Character Consistency Evaluation, Hard-Fail Safety Gating, Evaluation Infrastructure & Frameworks, Scalable Oversight, Replay-Based Evaluation, Behavioral Taxonomies, Semantic Auditing, Epistemic Provenance, CUSUM Control Charts, Fixed Harness / Mutation Budget Architecture, Evaluation Gating, Adversarial Robustness, AI Alignment, Frontier AI Safety, Mechanistic Interpretability, Latent Evaluation Awareness, Activation Verbalization, Constitutional AI, 4x4 Refusal Matrix

**Measurement & Research Methods:** Psychometrics, Construct Validity, Criterion Validity, Convergent & Discriminant Validity, Test-Retest Reliability, Inter-Rater Reliability, Uncertainty Quantification, Experimental Design, A/B Testing, Causal Inference, Incrementality Testing, Lift Studies, Power Analysis, Source Independence Analysis, Ablation Studies, Behavioral Typologies, Item Response Theory, Classical Test Theory, Bayesian Uncertainty, Late Fusion Architecture, LIWC Methodology, Computational Linguistics, Psycholinguistics, HDBSCAN Clustering, BERTopic, Prophet Time-Series, NetworkX Co-occurrence, Minimum Detectable Effect (MDE), Difference-in-Differences, Within-Subjects Designs, Preregistration, Offline Replay Experiments, FDR-Controlled Subgroup Analysis, Break-Even and ROI Modeling, Factor Analysis, Cohen's Kappa, PABAK, Adverse-Impact Analysis, LIWC, Hand-Coded Content Analysis, Text Classification, LDA Topic Modeling, Collocation Analysis (logDice, nMPI), Discourse Analysis, Co-occurrence Network Analysis

**Behavioral Science & Qualitative Methods:** Pathway to Violence, Warning-Behavior Typologies, Habit Formation, Choice Friction in Interaction Design, Hick's Law, Jakob's Law, von Restorff Effect, Irony Process Theory, Kohlberg Moral Development, Semi-Structured Interviews, User Research, Behavioral Observation, Clinical Assessment, Forensic Assessment, Structured Threat Assessment

**Machine Learning & Modeling:** LLMs, Generative AI, Prompt Engineering, RLHF, Reward Modeling, Reinforcement Learning, CMDPs, Weak Supervision, Positive-Unlabeled Learning, Learning to Rank, Knowledge Graphs, Graph ML, Entity Resolution, Neuro-Symbolic AI, XGBoost, Logistic Regression, Survival Analysis, Semantic Clustering, NLP, Topic Modeling, Sentiment Analysis, Synthetic Data Generation, SHAP Explainability, Isotonic Calibration, MediaPipe, Whisper

**MLOps, Platform & Infrastructure:** Model Context Protocol (MCP), End-to-End ML Pipeline Architecture, Model Lifecycle Management, Distributed Computing, Apache Spark, PySpark, Databricks, Delta Lake, AWS SageMaker, AWS Glue, AWS EMR, Airflow, Docker, Kubernetes, CI/CD, DuckDB, PyArrow

**Governance, Safety & Compliance:** Responsible AI, AI Governance, ISO/IEC 42001, NIST AI RMF, EU AI Act, FCRA Compliance, Bias Mitigation, Algorithmic Auditing, Privacy-Preserving ML, RBAC, Trust and Safety, Content Moderation, Platform Integrity, Over-Enforcement Measurement

**Technical Stack:** Python, PySpark, SQL, R, SAS, SPSS, Excel, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, Scala, Kafka, Hive, Tableau, Power BI, Databricks MLflow, AWS ML Services, HTML, CSS, JavaScript, Astro SSG, Firebase